

Deep Research Report: Metacognitive Continual Learning for Non-Stationary Smart Grid Dynamics

Executive Summary of the Project

The three uploaded files describe a tightly integrated PhD research program titled "Metacognitive Continual Learning for Non-Stationary Smart Grid Dynamics." The `ROADMAP.md` outlines an 18-month implementation plan, while `Topic_1_Data_Augmentation.py` and `Topic_1_Simulation.py` together provide a working prototype: the first synthesizes seasonal distribution shifts from a base smart-grid dataset, and the second runs Optimal Power Flow on the IEEE 30-bus system and trains an LSTM voltage predictor under a novel EWC-IXER (Elastic Weight Consolidation + Importance-Weighted Experience Replay) algorithm. The research aspires to publish at NeurIPS/ICML and IEEE Transactions on Smart Grid, with KPIs of >40% forgetting reduction, sub-100 ms GPU inference, and calibration error <0.05.

This report consolidates (i) what the three files actually contain, (ii) the academic literature needed to ground each component, and (iii) concrete gaps you must close before the paper can be submitted.

1. Synthesis of the Uploaded Artifacts

1.1 Research Objective and Scope

The roadmap defines the goal as a metacognitive continual learning framework that enables AI agents to autonomously detect, diagnose, and adapt to distribution shifts in smart grids without catastrophic forgetting. The specific non-stationary phenomena targeted are seasonal cycles, equipment degradation, renewable expansion, and extreme weather. The system is positioned as Chapter 3 of a PhD thesis and is explicitly scoped against six baselines: Naïve Fine-Tuning, EWC, Experience Replay (ER), PackNet, Progressive Neural Networks (PNN), and the proposed EWC-IXER + Metacognition.

1.2 The Data-Augmentation Pipeline (Topic_1_Data_Augmentation.py)

The `SeasonalAugmenter` class transforms a base CSV (e.g., `smart_grid_dataset.csv`) into a multi-season dataset whose rows carry `season` and `task_id` labels. The exact, hard-coded seasonal physics are:

Season	Solar	Load	Temperature	Wind
Winter	$\times 0.4$ (−60%)	$\times 1.3$ (+30%)	−15 °C	$\times 1.2$ (+20%)
Summer	$\times 1.4$ (+40%)	$\times 1.15 + \text{Gaussian afternoon peak}$ $1 + 0.3 \cdot \exp(-(h-15)^2/10)$	+10 °C	$\times 0.9$ (−10%)
Spring	$\times 0.8 \cdot \mathcal{N}(1.0, 0.15)$	$\text{base} \cdot \mathcal{N}(1.0, 0.15)$	$\text{base} + \mathcal{N}(0, 5)$	$\text{base} \cdot \mathcal{U}(0.7, 1.3)$
Fall	$\times 0.7$ (−30%)	$\times 1.05$ (+5%)	−5 °C	$\times 1.15$ (+15%)

Change-point detection is implemented via a sliding-window Levene + independent t-test with $p < 0.01$ thresholds, window size $\min(100, n//10)$, step $\text{window}/2$, on the `Power_Consumption` column. This is a frequentist surrogate for true Bayesian Online Change Point Detection (BOCPD) and will need to be replaced (see §3.5) before submission.

1.3 The Simulation Pipeline

(Topic_1_Simulation.py)

The script wires three components:

1. PyPower / IEEE 30-bus OPF generates ground-truth voltage magnitudes for each seasonal sample via `runopf(case30, ...)`, updating loads at buses 2, 4, 6, 7, 11, 14, 16, 18, 19, 22, 23, 28 and adjusting generation for solar/wind.
2. **VoltagePredictorLSTM** — a 2-layer LSTM with `input_dim=10`, `hidden_dim=64`, `num_layers=2`, `output_dim=30`, terminating in a linear head that maps the last-step hidden state to all 30 bus voltages.
3. **EWCIXERAgent** — a continual-learning trainer combining EWC and a priority-weighted replay buffer (default size 1000), with total loss
$$\mathcal{L}_{\text{total}} = \mathcal{L} + \lambda_{\text{EWC}} \sum_{\tau, i} F_i^{(\tau)} (\theta_i - \theta_i^{(\tau)})^2 + 0.5 \cdot \mathcal{L}_{\text{replay}}$$
 with `lambda_ewc=1000`, MSE task loss, Adam optimizer, and 20 epochs/task. Buffer sample priority is `task_loss.item()` and replacement is probabilistic with inverse priority. Reported metrics are Average Accuracy (defined as $1/(1+\text{RMSE})$) and Backward Transfer*, both computed by an external `metrics` module.

1.4 Critical Gaps Detected in the Code (Compared to the Roadmap's Promises)

The roadmap promises a metacognitive framework, but the present code implements only the EWC-IXER substrate. The following components are named but not yet coded:

- No MC-Dropout, ensembles, or Bayesian layers in **VoltagePredictorLSTM**, despite the uncertainty-quantification claim. Fisher information and loss-priority replay are the only forms of "importance" present.
- No conformal prediction integration (no `MapieRegressor` calls).
- No self-diagnosis module that distinguishes feature drift vs. gradual shift.
- No Forward Transfer (FWT) metric — only BWT is computed.

- The "Bayesian" change point detection is frequentist: there is no hazard rate, no run-length posterior, no recursive Bayesian update.
- Voltage error reported is RMSE (via `metrics_ps.rmse`), not MAE, despite the roadmap mentioning both.

These gaps are the largest blockers between the prototype and a publishable NeurIPS/ICML paper.

2. Literature You Must Cite to Write This Paper

2.1 Foundational Continual Learning (Pillar 1)

Elastic Weight Consolidation (EWC) — Kirkpatrick et al., "Overcoming catastrophic forgetting in neural networks," PNAS 114(13), 2017. The canonical reference for the Fisher-Information-Matrix penalty $\sum_i F_i(\theta_i - \theta_i^*)^2$ that anchors the EWC half of EWC-IXER [PNAS](#). EWC is described as "an elegant approach to selective regularization of network parameters when switching tasks. It uses the Fisher Information Matrix (FIM) to identify" critical weights [arXiv 1802.02950](#), and a synaptic-consolidation analogy is used to motivate the elastic anchoring [Rylan Schaeffer notes](#).

PackNet — Mallya & Lazebnik, "PackNet: Adding Multiple Tasks to a Single Network by Iterative Pruning," CVPR 2018 [arXiv 1711.05769](#) [CVPR 2018 PDF](#). It "assigns parameter subsets to tasks via iterative pruning" [ResearchGate](#), packing tasks into one network by iteratively pruning, freezing, and retraining at each task change [Continual World](#). For your benchmark, the roadmap retains 60 % of weights for old tasks.

Progressive Neural Networks (PNN) — Rusu et al., 2016 [arXiv 1606.04671](#). PNNs are "immune to forgetting and can leverage prior knowledge via lateral connections" between task-specific columns [arXiv PDF](#), enabling "deep compositionality" but at the cost of growing parameter count per task. This is the natural ceiling baseline for your forgetting metric.

Experience Replay — Rolnick et al., "Experience Replay for Continual Learning," NeurIPS 2019 [NeurIPS PDF](#). Once the buffer is full, "we use reservoir sampling to decide when to replace elements of the buffer with new experiences" — the exact strategy your IXER buffer extends with priority weights. Stratified variants by class/event type are documented in [Stratified Experience Replay](#) and [PMC review](#), which justifies your 30/30/40 split (faults / rare / nominal). Recent improvements to dark experience replay and reservoir sampling appear in Improvements of Dark Experience Replay and Reservoir Sampling (2025) [arXiv 2504.20932](#).

Task-Free CL & Memory-Aware Synapses — Aljundi et al., "Task-Free Continual Learning," CVPR 2019 [arXiv 1812.03596 CVPR PDF](#). Crucial because your BOCPD-driven discovery converts the smart-grid stream into an online, boundary-free problem — exactly the regime MAS was designed for. Recent extensions include online task-free CL via discrepancy mechanism (KBS 2025) [ScienceDirect](#).

Meta-Learning / MAML — Finn, Abbeel, Levine, "Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks," ICML 2017 [arXiv 1703.03400 PMLR](#). Cited by the roadmap as the few-shot baseline for Month 6. Javed & White, 2019 (OML) is its continual-learning counterpart, learning representations resilient to interference.

Modern surveys you should cite for the introduction. The 2024 arXiv survey "Continual Learning and Catastrophic Forgetting" provides an updated taxonomy [arXiv 2403.05175](#), and the 2025 Energy journal paper "Continual learning for energy management systems: A review" is a domain-specific motivator emphasizing power systems, NILM and smart grids [ScienceDirect](#).

2.2 Continual-Learning Evaluation Metrics

For BWT/FWT/Average Accuracy you should cite the original GEM paper — Lopez-Paz & Ranzato, "Gradient Episodic Memory for Continual Learning," NeurIPS 2017 [arXiv 1706.08840](#). It defines "GEM minimizes backward transfer, while exhibiting negligible or positive forward transfer." A practitioner-friendly recap: The Metrics of Continual Learning [TDS](#) confirms "Average accuracy (Acc) measures the test performance, backward transfer (BWT) measures catastrophic forgetting, and forward transfer (FWT)...". Díaz-Rodríguez et al., 2018, "Don't forget, there is more than forgetting: new metrics for Continual

Learning," [arXiv 1810.13166](#) supplies the additional metrics (Task Interference Index, memory cost, etc.) the roadmap explicitly mentions.

The Avalanche library — Lomonaco et al., 2021, "Avalanche: an End-to-End Library for Continual Learning" [JMLR PDF IEEE](#) — is the de facto framework you should adopt; it provides "popular benchmarks (like SplitMNIST...)" along with strategies, metrics and loggers in five modules [GitHub docs](#).

2.3 Bayesian Online Change-Point Detection (Pillar 2)

The seminal reference is Adams & MacKay, 2007, "Bayesian Online Changepoint Detection," [arXiv 0710.3742 Princeton PDF](#). They present "a Bayesian changepoint detection algorithm for online inference. Rather than retrospective segmentation, we focus on causal prediction" — exactly the streaming requirement of a real grid. Gundersen's 2019 tutorial gives a clean derivation of the run-length posterior [Gundersen blog](#), and a UAI 2020 extension predicts the time until next change point, which is a natural extension for grid horizon planning [UAI 2020 PDF](#). Open-source implementations exist at [dtolpin/bocd](#) and Meta's [Kats](#), either of which would replace your current Levene/t-test scaffold and make the "Bayesian" claim defensible.

2.4 Uncertainty Quantification & Metacognitive AI (Pillar 3)

Conformal prediction — Angelopoulos & Bates, "A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification," [arXiv 2107.07511](#); the published Foundations and Trends monograph supplies the theoretical guarantees for 95 % coverage [Emerald](#). The companion repo [aangelopoulos/conformal-prediction](#) is "the easiest way to start using conformal prediction (a.k.a. conformal inference) on real data."

Adaptive Conformal Inference for time series — Zaffran, Féron et al., 2022, "Adaptive Conformal Predictions for Time Series," ICML [arXiv 2202.07282](#), which delivers "efficient prediction intervals for day-ahead forecasting" — the precise use case for your voltage forecasts. The MAPIE library implements this via `MapieTimeSeriesRegressor / TimeSeriesRegressor` [MAPIE TS tutorial API ref](#); ACI "allows you to adapt the conformal inference (i.e., the quantile)" so that

interval width grows when coverage drops — the natural drift-detection signal $\eta(t)$ in your roadmap. A practitioner reference is Cordier et al., 2023, "Flexible and Systematic Uncertainty Estimation with Conformal Prediction" [PMLR](#).

Metacognitive AI. The Cambridge volume Metacognitive Artificial Intelligence (2024), Chapter 9 "Uncertainty Quantification's Role in Metacognition," is the closest book-length treatment of the framing you adopt: "Metacognitive AI represents an approach to design agents that continue safely learning and adapting as they encounter new or uncertain scenarios" [Cambridge](#). For survey coverage of UQ in deep nets, Abdar et al., 2021 and the 2022 review "A unified review of uncertainty quantification in deep learning" [arXiv 2210.03736](#) decompose epistemic vs. aleatoric uncertainty — the decomposition the roadmap promises but the code does not yet implement. A practical 2025 review of UQ for neural-network potentials in npj Computational Materials [Nature](#) is a useful template for how UQ is reported in a domain-science venue. You should also point to the NeurIPS-aligned Workshop on Metacognition in Generative AI call to position your work within that growing community [Metacognition GenAI](#).

2.5 Smart-Grid AI under Non-Stationarity (Pillar 4 — the application story)

For the introduction's "why this matters" narrative, three sources stand out:

- The 2025 Applied Energy paper "Continual learning for energy management systems: A review of methods and applications" explicitly maps CL to power systems, NILM and smart grids [ScienceDirect](#) — your single best citation to claim a research gap.
- "Load forecasting under distribution shift: An online quantile mixture-of-experts" (Applied Energy, 2025) demonstrates "extensive experiments across diverse distribution shift scenarios" for grid load forecasting [ScienceDirect](#), giving you a direct competitor to benchmark against.
- Zhang et al., 2021, "Time Series Anomaly Detection for Smart Grids: A Survey" [arXiv 2107.08835](#) emphasizes that "real-world time series are often non-stationary" — perfect motivation for the BOCPD component.

For drift-adaptive LSTMs in load forecasting, the cleanest precedent is DA-LSTM — "A dynamic drift-adaptive Long Short-Term Memory framework for interval

load forecasting," Engineering Applications of AI, 2023 [ScienceDirect](#) [arXiv 2305.08767](#), which uses concept-drift detection to retrain on the fly. LSTMDD (PMC 2024) generalizes the idea to cloud-resource forecasting [PMC](#), and a recent Energies MDPI study confirms LSTM is "particularly effective in tasks like predicting voltage trends in power grids, where sudden changes and long-term patterns are" co-present [MDPI Energies 17/22/5528](#). A 2024 SSRN paper, "An Adaptive Ensemble-LSTM for Enhanced Load Forecasting," addresses concept drift via ensembles — a strong baseline candidate [SSRN PDF](#).

For renewable forecasting under distribution shift, the 2025 Springer Machine learning for renewable energy forecasting review establishes that "DNN-based STEF has become dominant for highly nonlinear settings and multivariate" data [Springer](#); Nature Scientific Reports 2025 on explainable AI for green energy forecasting [Nature](#) supplies a recent XAI-energy citation. A standard benchmark dataset is PSML — Zheng et al., Nature Scientific Data, 2022, "A multi-scale time-series dataset with benchmark for machine-learning in decarbonized energy grids" [Nature](#), which complements your synthetic seasonal augmentation with real multi-scale data.

For the synthetic-time-series augmentation half of your method, cite Turowski et al., "Generating synthetic energy time series: A review" (Renewable & Sustainable Energy Reviews, 2024) [ScienceDirect](#) and Heidrich et al., 2023, "Controlling non-stationarity and periodicities in time series generation using conditional invertible neural networks" [Springer](#) — these justify your hand-crafted seasonal multipliers as a first-step augmentation strategy that future work could replace with cINN-based generative models.

2.6 IEEE 30-Bus & PyPower Optimal Power Flow (Pillar 5 — the testbed)

The IEEE 30-bus is the field-standard benchmark; for OPF formulation cite Babu & Mohan, "Optimal Power Flow Analysis of IEEE-30 bus System using Genetic Algorithm" [IJOER PDF](#). The most relevant machine-learning-meets-OPF precedent for your work is the Stanford CS229 study "Predicting Solutions to the Optimal Power Flow Problem," which used "the IEEE 30 bus network as our static network model" with MATPOWER ground truth [Stanford PDF](#) — almost the same setup as your `Topic_1_Simulation.py`. Babatunde et al., 2024 on

MEBO-based OPF for IEEE 30/118-bus [ResearchGate](#) and the 2024 IEEE Access paper on stochastic OPF integrating wind/PV with the IEEE 30-bus [IEEE 10589639](#) provide the renewable-aware OPF context. For benchmark hygiene cite the Power Grid Library task-force report [PGLib OPF](#), which standardizes the AC-OPF instances. A 2025 IEEE paper, "An Explainable AI Framework for AC Optimal Power Flow," is a direct intellectual neighbor: it "proposes an interpretable deep learning framework to address the AC-OPF problem, offering explainability without sacrificing accuracy" [IEEE 11177141](#).

2.7 Explainability for the Grid (Pillar 6)

For SHAP vs. Integrated Gradients, the canonical contrast is the Fiddler review "Integrated Gradients vs SHAP" [Fiddler](#). For application-grade evidence in energy, Sharbati 2026 "An interpretable XGBoost-SHAP framework for forecasting [grid] capacity" [ScienceDirect](#) and the IRJMETS paper integrating SHAP + LIME for power systems [IRJMETS PDF](#) are useful. The 2025 Nature Scientific Reports on XAI in green-energy forecasting [Nature](#) and a 2025 arXiv on "Explaining Deep Learning-based Anomaly Detection in [energy] data" [arXiv 2501.06099](#) are recent and citable. The `shap.GradientExplainer` documentation [SHAP docs](#) describes the unification with integrated gradients you'd actually use in code.

3. What Is Still Needed to Write the Paper

The following sub-sections enumerate the missing scientific and engineering deliverables, framed as a paper-completion checklist.

3.1 Methodological Novelty: Sharpen the EWC-IXER Contribution

EWC and ER are individually 8-year-old methods; reviewers will press you on novelty. The paper's contribution must be defended on at least three axes:

1. The "I" in IXER — formalize Importance-Weighted Experience Replay as a buffer-management policy whose priority p_i is jointly a function of (a) loss-gradient magnitude (epistemic difficulty), (b) BOCPD posterior run-length (rarity), and (c) physics-informed event tags (faults / overloads). Currently the code uses only loss magnitude, which collapses to a vanilla prioritized buffer.
2. The metacognitive coupling — the system must modulate λ_{EWC} and the replay/task-loss ratio in response to the conformal-coverage drift signal $\eta(t)$. This closes the loop and is genuinely novel; right now `lambda_ewc=1000` is fixed.
3. Theoretical guarantee — borrow from Adaptive Conformal Inference (Zaffran et al.) to provide a long-run coverage guarantee under distribution shift, then show empirically that the metacognitive trigger preserves this guarantee while EWC-IXER preserves task accuracy. NeurIPS-tier reviewers will expect at least this much theory.

3.2 Implement the Missing Uncertainty Layer

The code provides no MC-Dropout, no ensemble, no Bayesian layer. A minimal viable version is:

- Add `nn.Dropout(p=0.2)` between the LSTM and the FC head, sample 30 stochastic forward passes at inference, and report mean $\pm$ std as epistemic uncertainty.
- Wrap the predictor in a `MapieTimeSeriesRegressor` with the ACI strategy [MAPIE TS tutorial](#); compute rolling 95 % coverage, and trigger task-head creation when coverage falls below 80 % for ≥ 30 minutes (matching the roadmap's 15 % violation threshold).

- Decompose total predictive variance into aleatoric (heteroscedastic head, $\sigma^2(x)$) and epistemic (MC variance) components, following the 2022 review [arXiv 2210.03736](#).

3.3 Replace the Frequentist Change-Point Hack with True BOCPD

Re-implement `bayesian_changepoint_detection` using the Adams–MacKay run-length posterior with a constant hazard rate $H(r)=1/\lambda$ (try $\lambda=144$ for 10-min-resolution daily-period grid data) and a Normal-Inverse-Gamma conjugate prior. Reference implementations are at [dtolpin/bocd](#) and Meta's [Kats bocpd](#). Without this, calling the system "Bayesian" is indefensible.

3.4 Add Forward Transfer & Task Interference Metrics

The current simulation only computes Average Accuracy and BWT. Add:

- FWT — accuracy on task t before any training on it, minus a chance-level reference (Lopez-Paz & Ranzato, 2017 [arXiv 1706.08840](#)).
- Task Interference Index — pairwise gradient-cosine drop between consecutive tasks (Díaz-Rodríguez et al., 2018 [arXiv 1810.13166](#)).
- Voltage MAE in addition to RMSE — RMSE alone is sensitive to outliers; reviewers in IEEE Transactions on Smart Grid expect both, plus per-bus error heatmaps.
- Calibration Error — Expected Calibration Error (ECE) on the conformal intervals; the roadmap KPI is < 0.05 .

3.5 Strengthen the Experimental Protocol

The roadmap promises 10 random seeds and Wilcoxon signed-rank tests; the code does neither. Wrap the simulator in Avalanche [JMLR PDF](#) so that all six baselines (Naïve / EWC / ER / PackNet / PNN / EWC-IXER+Meta) share an identical

training/eval loop, identical buffers (where applicable), and identical seeds. Then:

- Run $10 \text{ seeds} \times 6 \text{ methods} \times 6 \text{ tasks} = 360$ training runs.
- Report mean $\pm 95\%$ CI for ACC, BWT, FWT, ECE, latency, peak memory.
- Wilcoxon at $\alpha=0.05$ with Bonferroni correction for the 5 pairwise comparisons against your method.
- Add a real-data validation on PSML [Nature Sci Data](#), not just the synthetically augmented 7-day base set, otherwise the seasonal multipliers are circular evidence.

3.6 Edge-Deployment Profiling

The KPI of "sub-500 ms on edge devices" requires Jetson Nano benchmarks. Use TensorRT FP16 conversion for the LSTM, profile latency at batch=1 and batch=32, and report energy per inference (mWh). The IEEE PES audience will weight this heavily; the NeurIPS audience will not, but a single-page appendix is sufficient.

3.7 Ablations the Reviewers Will Demand

At minimum:

1. EWC only vs ER only vs EWC+ER vs EWC+ER+Metacognitive trigger — to isolate the value of the metacognitive component (probably 5–15 points of forgetting reduction).
2. Buffer composition ablation — vary the 30/30/40 split to {20/30/50, 40/40/20, uniform}.
3. $\lambda_{\text{EWC}} \in \{10, 100, 1000, 10000\}$ — show your value is not cherry-picked.
4. Adaptive vs. fixed λ — adaptive driven by conformal coverage should win, otherwise the metacognitive story collapses.

3.8 Threats to Validity to Pre-Empt in the Discussion

- Synthetic seasonality is too clean — real grid distribution shift is non-stationary at multiple scales; cite Zhang 2021 [arXiv 2107.08835](#) and Turowski 2024 [ScienceDirect](#) and explicitly limit the claim to "controlled distribution shift."
 - IEEE 30-bus is small — promise IEEE 118-bus and a real distribution feeder in Phase 3 (Months 15-16 of the roadmap).
 - EWC's Fisher diagonal is a known approximation — reviewers will flag the off-diagonal terms; cite the modern critique of EWC variants in [arXiv 1802.02950](#) and acknowledge it.
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4. Recommended Paper Outline (NeurIPS-Style 9 Pages + Appendix)

Section 1 · Introduction. Lead with the smart-grid motivation (Continual Learning for Energy Management Systems review, Applied Energy 2025 [ScienceDirect](#)). Pose three questions: (Q1) Can an agent autonomously detect when the grid has shifted? (Q2) Can it learn the new regime without forgetting the old? (Q3) Can it know when it doesn't know?

Section 2 · Related Work. Three subsections: 2.1 Continual learning (Kirkpatrick 2017, Mallya 2018, Rusu 2016, Aljundi 2019, Finn 2017, Rolnick 2019); 2.2 Uncertainty quantification & metacognition (Angelopoulos & Bates 2021, Zaffran 2022, Cambridge 2024 vol. Ch. 9); 2.3 ML for power systems (DA-LSTM 2023, distribution-shift load forecasting 2025, PSML 2022).

Section 3 · Problem Formulation. Define the smart-grid CL problem as a stream $\{(x_t, y_t)\}$ with unknown task boundaries $\{T_k\}$, voltage prediction targets from PyPower OPF on IEEE 30-bus, and a coverage constraint $P(y_t \in C_{\alpha}(x_t)) \geq 1 - \alpha$.

Section 4 · Method: EWC-IXER + Metacognitive Adaptation. 4.1 BOCPD task discovery (Adams & MacKay 2007); 4.2 Importance-weighted replay buffer (extending Rolnick 2019 with priority by gradient $\times$ run-length $\times$ event tag); 4.3

Adaptive EWC penalty modulated by ACI coverage (Zaffran 2022); 4.4 Self-diagnosis decision rule (new task head vs. incremental update).

Section 5 · Experiments. 5.1 Datasets (synthetic seasonal augmentation + PSML real data); 5.2 Baselines (the six listed); 5.3 Metrics (ACC, BWT, FWT, ECE, latency); 5.4 Main results with significance tests; 5.5 Ablations (§ 3.7).

Section 6 · Analysis & Discussion. Per-task per-bus error maps; SHAP attribution drift across tasks (Fiddler 2021 framing [Fiddler](#)); Limitations (§ 3.8); Broader impacts on grid resilience.

Section 7 · Conclusion. Re-state KPIs achieved; tease the Nature Energy / Joule extension on climate-projection scenarios.

Appendix. Edge-deployment numbers; full hyperparameter tables; code availability statement.

5. Concrete Next-Step Plan (Aligned to the Roadmap)

Week	Action	Output
1	Adopt Avalanche; reimplement all six baselines under one harness	Baseline parity report
2	Replace frequentist CPD with Adams–MacKay BOCPD via Kats	Task-discovery validation curves
3	Add MC-Dropout + MAPIE ACI to <code>VoltagePredictorLSTM</code>	Coverage-vs-time plots
4	Implement adaptive λ_{EWC} tied to coverage signal	Closed-loop metacognition
5	Run full 10-seed × 6-method experiments	Wilcoxon-tested results table
6	Add MAE, FWT, ECE, Task Interference Index	Complete metric panel
7	PSML real-data validation	External-validity figure
8	Jetson Nano profiling	Latency/energy table
9	First full draft (Methods + Experiments)	Paper v0.1
10	SHAP / Integrated Gradients drift analysis	Explainability section
11–12	Advisor revisions + abstract polishing	NeurIPS submission

6. Bottom Line

Your three files form a coherent and ambitious project, but at the moment the prototype delivers only the EWC + replay layer of a five-layer stack (CL $\leftarrow$ BOCPD $\leftarrow$ ACI $\leftarrow$ self-diagnosis $\leftarrow$ metacognitive control). Closing those gaps — particularly (a) replacing the Levene/t-test scaffold with true BOCPD, (b) integrating MAPIE adaptive conformal inference, (c) coupling λ_{EWC} to the coverage signal, and (d) running a 10-seed Avalanche-based benchmark against six baselines on both synthetic and PSML real data — is what stands between the current code and a NeurIPS / IEEE Transactions on Smart Grid submission. The literature surveyed above (Kirkpatrick 2017, Mallya 2018, Rusu 2016, Rolnick 2019, Aljundi 2019, Finn 2017, Adams & MacKay 2007, Angelopoulos & Bates 2021, Zaffran 2022, Lopez-Paz & Ranzato 2017, Lomonaco et al. 2021, plus the 2024–2025 energy-CL and distribution-shift forecasting reviews) covers every citation you need for the introduction, related work, methods, and discussion sections of that paper.